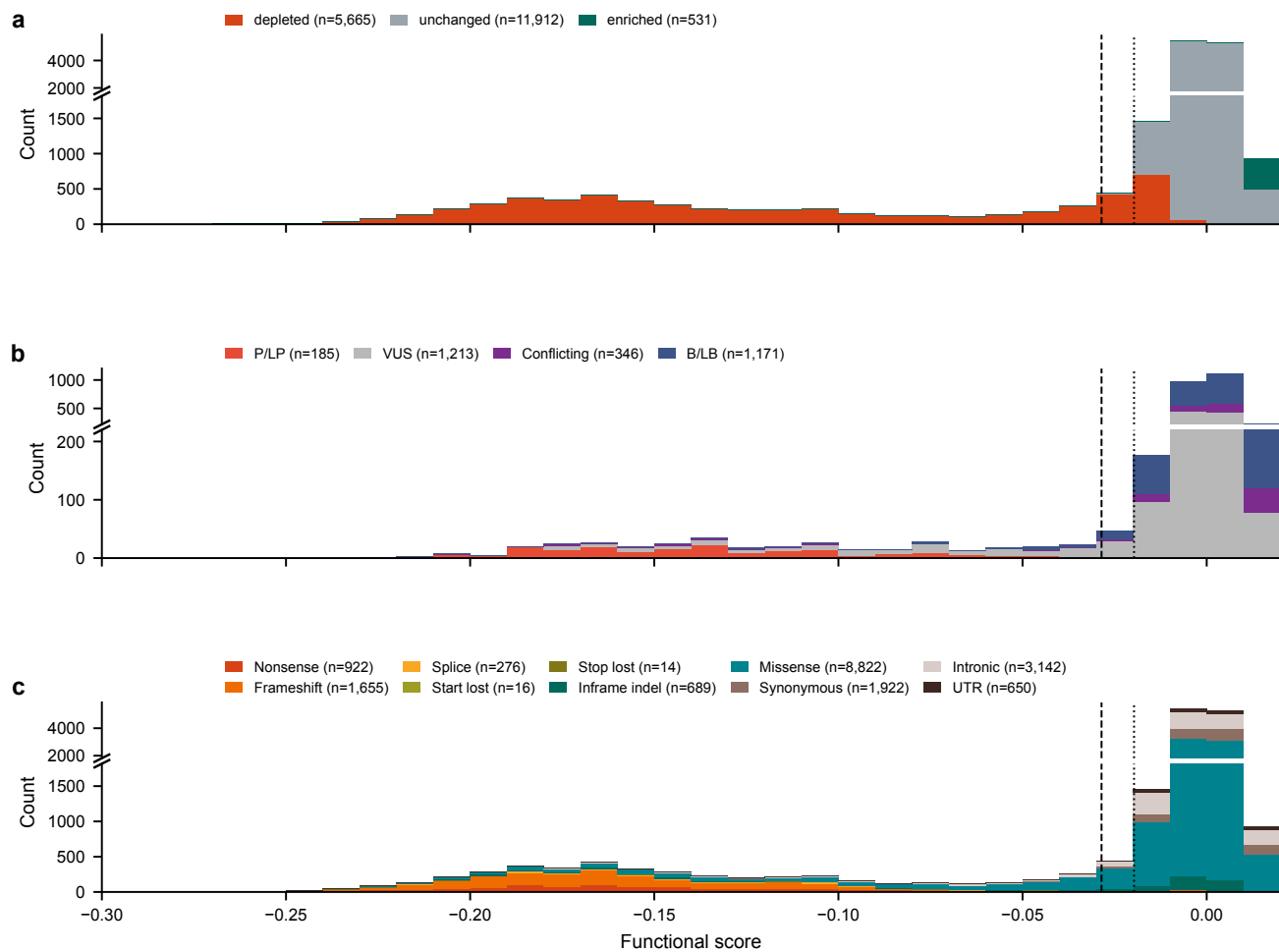




d Synonymous (benign) (n=1,922)

Depleted (LoF)	51	16	56
Unchanged	0	1	1,699
Enriched (GoF)	0	0	99

f ClinVar B/LB (benign) (n=1,171)

	25	13	35
	0	0	1,008
	0	0	90

h KURIS B/LB miss. (benign) (n=28)

	0	0	1
	0	0	23
	0	0	4

e Truncating (path) (n=2,853)

Depleted (LoF)	2,778	9	10
Unchanged	1	1	54
Enriched (GoF)	0	0	0

Abnormal Indeterminate Normal

g ClinVar P/LP (path) (n=185)

	184	0	0
	0	0	1
	0	0	0

Abnormal Indeterminate Normal

i KURIS/NDD miss. (path) (n=17)

	14	0	1
	0	0	2
	0	0	0

Abnormal Indeterminate Normal

Supplementary Figure 2. Comparing Waters classifications based on directionality and q-values to the GMM score-cutoff-derived classification.

(a–c) Functional-score histograms on a split count axis, stratified by (a) Waters functional class, (b) ClinVar classification, and (c) molecular consequence; the abnormal (dashed) and normal (dotted) score cutoffs are overlaid. (d–i) Cross-tabulation of Waters functional class against GMM class for benign control sets (d, synonymous; f, ClinVar benign/likely-benign; h, KURIS benign/likely-benign missense) and pathogenic control sets (e, truncating; g, ClinVar pathogenic/likely-pathogenic; i, KURIS/NDD missense). In (d–i), the count is printed in each cell and the cell shading is proportional to that cell's fraction of the control set.